Investigating the Effects of Coffee Type and Brand on the Acidicity of Coffee

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STAT404 Final Project

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Formulation

- **Objective**
  - Our primary objective is to optimize for the lowest acidity (highest pH value) across coffee type, brand, and temperature..
  - This is practically important because low-acid coffee reduces stomach and dental irritation while providing a smoother flavor
- **Treatments Factors**:
  - Brand of coffee (with levels of Starbucks and Tim Hortons)
  - Type of coffee (with levels of Americano and Latte).
- **Blocking Factor**: Temperature (with levels of iced and hot).
  - Temperature systematically affects the acidity (hot vs iced), we don't want it to be part of our research question
- **Experimental units**: One cup of medium-sized coffee prepared for a given brand-type combination at a specified temperature (hot or iced)
- **Measurement of the response**: The acidity of the coffee samples will be quantified by their pH value and will be measured using a digital pH meter. Lower pH values indicate higher acidity and vice versa.

Design

- **Factor levels**:

  All of our factors have 2 levels (see details in the previous slide).
- **Design Type**:

  Since all of our treatment and blocking factors have 2 levels and we are interested in estimating effects up to all factor interactions, we implemented a 2^3 **full factorial design** with two treatment factors and one blocking factor.
- **Randomization**:
  - Within each temperature block, we randomly permuted the four **Brand** $\times$ **Type** combinations using a random number generator and followed that order when preparing and measuring the coffees. Randomization controls for time-related confounds.
  - Randomization helps control for time-related confoundings, e.g., meter drifts.
- **Replication**:

  We do not have replication in this experiment.

- **Model Specification:**

$$Y_i = \beta_0 + \beta_1 r_{i1} + \beta_2 r_{i2} + \beta_3 b_i + \beta_{12} r_{i1} r_{i2} + \beta_{13} r_{i1} b_i + \beta_{23} r_{i2} b_i + \beta_{123} r_{i1} r_{i2} b_i + E_i, \quad (i = 1, \ldots, 8)$$

- **Parameter Estimation for the Regression Model:**

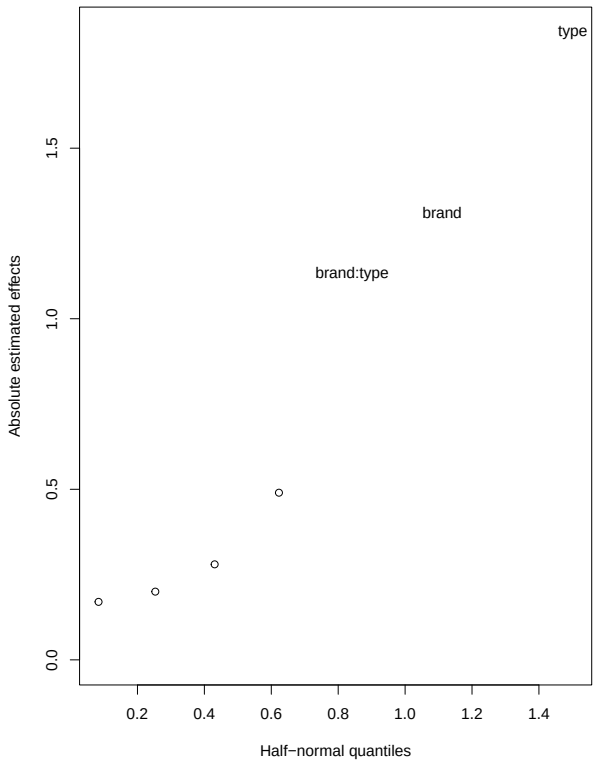
Effect	Estimate
Intercept	4.91
Brand (brand1)	1.31
Temperature (temp1)	−0.17
Type (type1)	1.84
Brand × Temperature (brand1:temp1)	−0.49
Brand × Type (brand1:type1)	−1.13
Temperature × Type (temp1:type1)	−0.20
Brand × Temperature × Type (brand1:temp1:type1)	0.28

- **Model Checking**:

  Because **all degrees of freedom are exhausted** in estimating the model parameters, no residual degrees of freedom remain, and thus Q–Q plots and Box–Cox transformations cannot be used for model diagnostics.

Analysis2 — Graphical Identification of the Significant Effects

- **Half-normal plot**

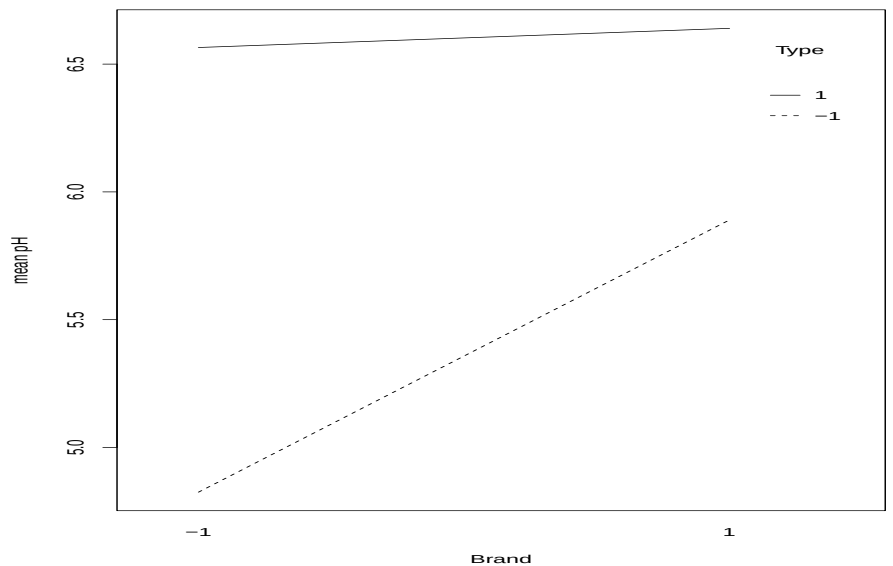


- **Interpretation**

  From the half-normal plot, we observed that the **main and interaction effects of the treatment factor** clearly stand out or deviate well above the line, indicating that they are likely to be statistically significant. However, the rest of the effects tend to lie close to the null distribution reference line, indicating that they are likely not to be significant.

Conclusion — Addressing Objective with Interaction Plots

- Since the interaction effect and the main effects of the treatment factors are very likely to be significant. We can assess the interaction plot and optimize for the combination of factors that gives the lowest acidity of coffee.
- **Interaction Plot** between treatment factors



- From the interaction plot, we see that the combination of **brand = 1 (Starbucks) and Type = 1 (Latte)** will give us the least acidity (highest pH), thus completing our objective.